Homework Assignment 1

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Rutgers University

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Problem 1:

Let X = Poisson(0.2), the number of typographical errors

(a)

$$P(X = 0) = e^{-0.2} \approx 0.819$$

(b)

$$\begin{aligned} P(X \geq 2) &= 1 - P(X = 0) - P(X = 1) \\ &= 1 - e^{-0.2}(1 + 0.2) \\ &\approx 0.018 \end{aligned}$$

Problem 2:

We have that $P(On) = \frac{1}{3}$ and $P(Off) = \frac{2}{3}$

- Case A: 3 examiners

$$\begin{aligned} P(Pass) &= P(Off)(\binom{3}{2}(0.4)^2(0.6) + \binom{3}{3}(0.4)^3) + P(On)(\binom{3}{2}(0.8)^2(0.2) + \binom{3}{3}(0.8)^3) \\ &= \frac{2}{3}(0.352) + \frac{1}{3}(0.896) \\ &\approx 0.533 \end{aligned}$$

- Case B: 5 examiners

$$\begin{aligned} P(Pass|On) &= \binom{5}{3}(0.8)^3(0.2)^2 + \binom{5}{4}(0.8)^4(0.2) + \binom{5}{5}(0.8)^5 \approx 0.942 \\ P(Pass|Off) &= \binom{5}{3}(0.4)^3(0.6)^2 + \binom{5}{4}(0.4)^4(0.6) + \binom{5}{5}(0.4)^5 \approx 0.317 \\ P(Pass) &= P(Pass|On)P(On) + P(Pass|Off)P(Off) \\ &\approx 0.525 \end{aligned}$$

He will have a higher chance of passing with 3 examiners.

Problem 3:

(a)

$$P(X=1) = F(1) - F(1^-) = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$
$$P(X=2) = F(2) - F(2^-) = \frac{11}{12} - \frac{3}{4} = \frac{1}{6}$$
$$P(X=3) = F(3) - F(3^-) = 1 - \frac{11}{12} = \frac{1}{12}$$

(b)

$$P(\frac{1}{2} < X < \frac{3}{2}) = F(\frac{3}{2}^-) - F(\frac{1}{2}) = \frac{1}{2}$$

Problem 4:

$$\begin{aligned} P(X=0|First) &= e^{-3} \\ P(X=0|Second) &= e^{-4.2} \\ P(X=0) &= P(X=0|First)P(First) + P(X=0|Second)P(Second) \\ &= 0.5e^{-3} + 0.5e^{-4.2} \approx 0.0324 \end{aligned}$$

Problem 5:

(a)

$$P(C_i) = \frac{2}{2n-1}$$

(b)

$$P(C_j \mid C_i) = \frac{2}{2n-3}$$